

Name: Rahul Gupta

Reg No: 2547242

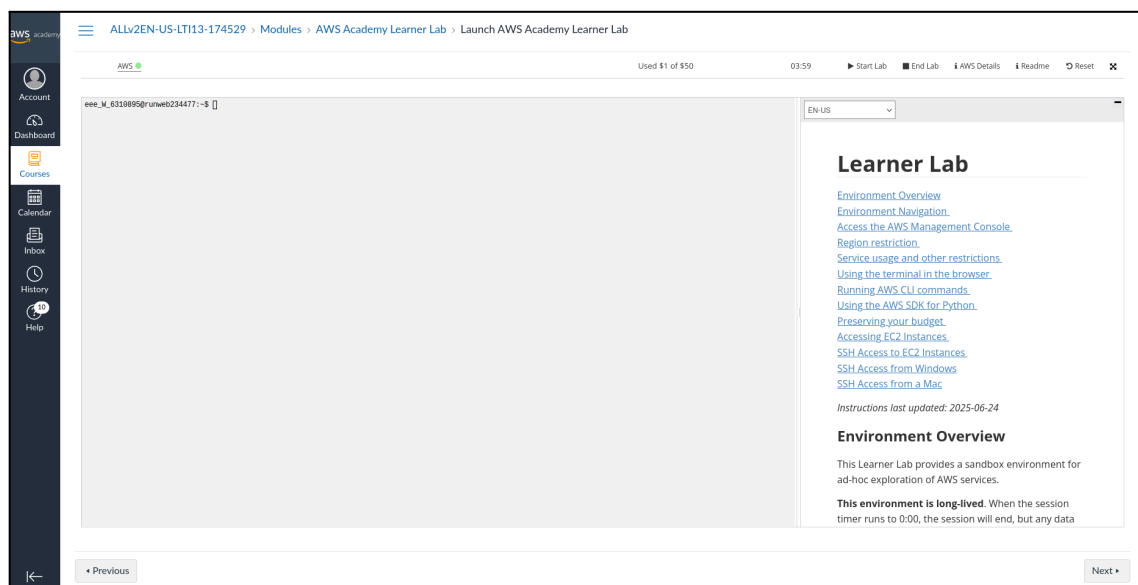
Class: 4 MCA B

Lab 5

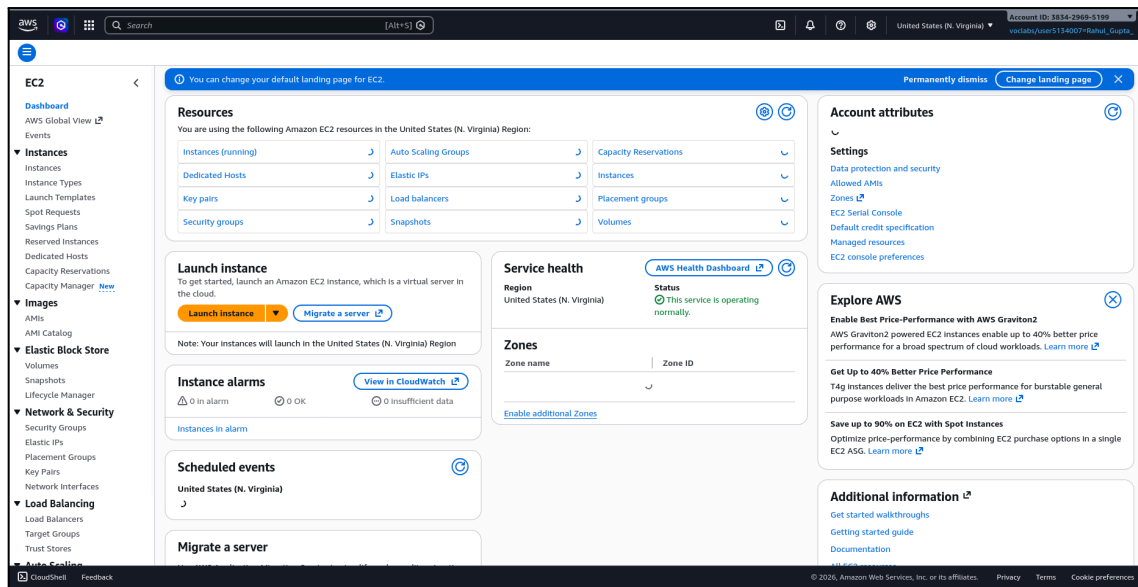
Aim: Assume that you have been appointed as a Cloud Engineer in an organisation that is planning to host applications on AWS. Your task is to provision and configure two different EC2 servers and demonstrate automated server initialisation using User Data.

Part 1: Launch and Configure the Ubuntu Server

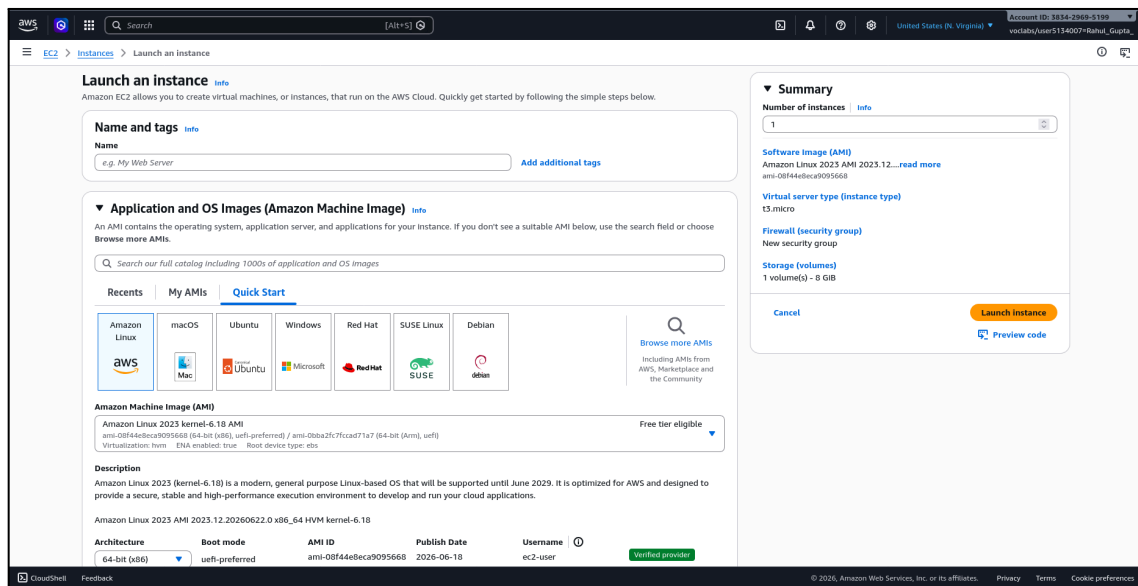
1. Open your AWS Learner Lab console, click on **Start Lab**, and once the icon turns green, click on **AWS** to open the AWS Management Console.



2. In the AWS Console search bar, type **EC2** and select the EC2 service from the dropdown.



3. Click on the **Launch instance** button on the EC2 dashboard.



4. Name your instance by typing **EC2-01** in the **Name** field.

Name and tags [Info](#)

Name

EC2-01

Add additional tags

5. Under **Application and OS Images (Amazon Machine Image)**, select **Ubuntu**. Ensure the default **Ubuntu Server 24.04 LTS (or 22.04 LTS) - Free Tier Eligible AMI** is selected.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

Canonical Ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 26.04 LTS (HVM), SSD Volume Type

Free tier eligible

ami-0b6d9d3d33ba97d99 (64-bit (x86)) / ami-0bc7f2dbdcc6b5303 (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 26.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 26.04, amd64 resolute image

Architecture

AMI ID

Publish Date

Username

Verified provider

64-bit (x86)

ami-0b6d9d3d33ba97d99

2026-06-04

ubuntu

6. Under **Instance type**, keep it as **t2.micro** (or the t3.micro default allowed by your Learner Lab).

▼ Instance type [Info](#) [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0116 USD per Hour On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

7. Under **Key pair (login)**, click the dropdown and select **vockey** (this is the pre-configured key pair provided by AWS Learner Lab).

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

vockey

▼

Create new key pair

8. Under **Network settings**, check the box that says **Create security group**.
- Check the box for **Allow SSH traffic from (Anywhere 0.0.0.0/0)** to ensure you can connect via terminal.
 - Check the box for **Allow HTTP traffic from the internet (0.0.0.0/0)** so your Apache web server can be accessed publicly.

▼ Network settings Info

Edit

Network Info

vpc-0033febc051601c6f

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

We'll create a new security group called 'launch-wizard-7' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere
0.0.0.0/0

☐ Allow HTTPS traffic from the Internet

To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the Internet

To set up an endpoint, for example when creating a web server

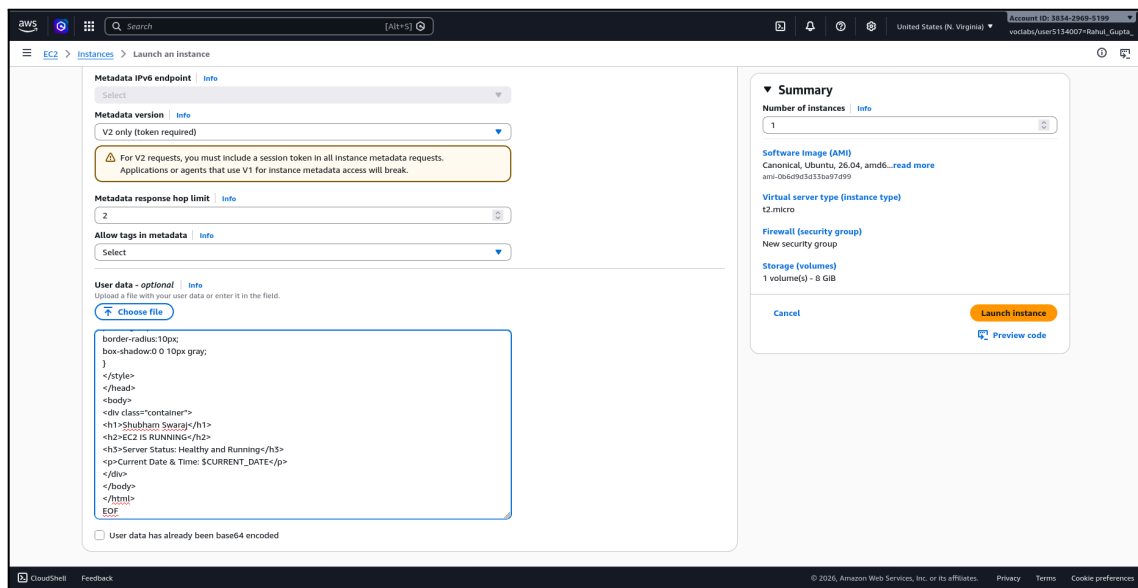
⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

×

9. Scroll down to the bottom of the page and click on **Advanced details** to expand the section.
- Scroll to the very bottom until you see the **User data** text box.
 - Copy the shell script provided below and paste it directly into the **User data** box

```
#!/bin/bash
apt update -y
apt upgrade -y
apt install apache2 -y
systemctl enable apache2
systemctl start apache2
CURRENT_DATE=$(date)
cat > /var/www/html/index.html <<EOF
<!DOCTYPE html>
<html>
<head>
<title>AWS Lab</title>
<style>
body{
font-family:Arial;
text-align:center;
background:#f5f5f5;
padding-top:100px;
}
.container{
background:white;
width:60%;
margin:auto;
padding:30px;
border-radius:10px;
box-shadow:0 0 10px gray;
}
</style>
</head>
<body>
```

```
<div class="container">
<h1>Shubham Swaraj</h1>
<h2>EC2 IS RUNNING</h2>
<h3>Server Status: Healthy and Running</h3>
<p>Current Date & Time: $CURRENT_DATE</p>
</div>
</body>
</html>
EOF
```



10. Click the **Launch instance** button on the right side panel.

▼ Summary

Number of instances | Info

1

Software Image (AMI)
Canonical, Ubuntu, 26.04, amd64...[read more](#)
ami-0b6d9d3d33ba97d99

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel

Launch instance

 Preview code

Success

Successfully initiated launch of instance (i-00e56130fd2578cb4)

► Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.

Manage detailed monitoring

Create Load Balancer

Create an application, network gateway or classic Elastic Load Balancer

Create Load Balancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Create AWS budget

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

Manage CloudWatch alarms

Disaster recovery for your instances

Recover the instances you just launched into a different Availability Zone or a different Region using AWS Elastic Disaster Recovery (DRS).

Disaster recovery for your instances

Monitor for suspicious runtime activities

Amazon GuardDuty enables you to continuously monitor for malicious runtime activity and unauthorized behavior, with near real-time visibility into on-host activities occurring across your Amazon EC2 workloads.

Monitor for suspicious runtime activities

Get instance screenshot

Capture a screenshot from the instance and view it as an image. This is useful for troubleshooting an unreachable instance.

Get instance screenshot

Get system log

View the instance's system log to troubleshoot issues.

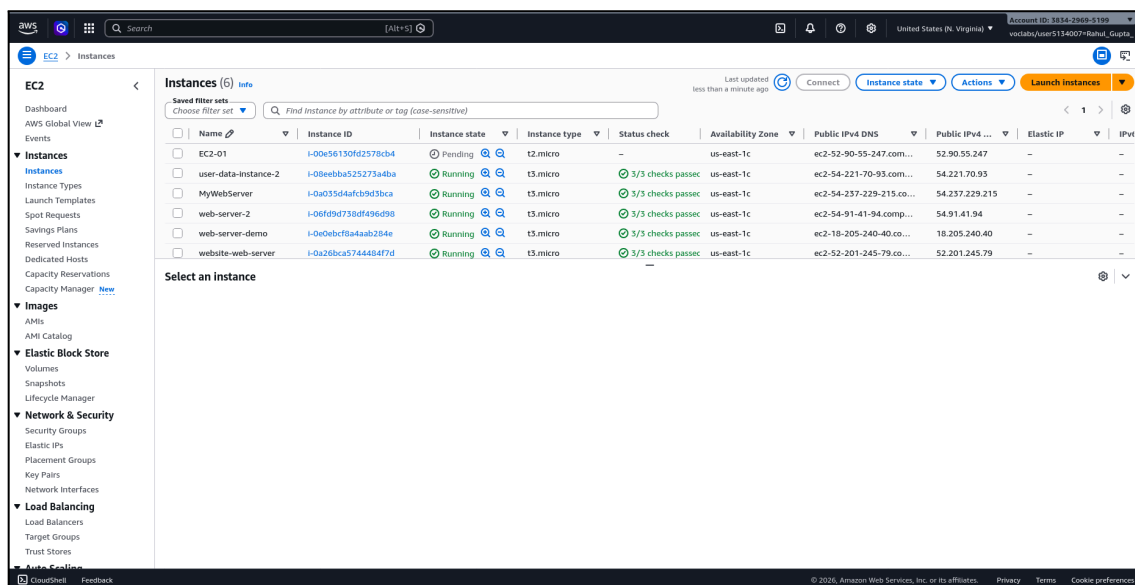
Get system log

CloudShell

Feedback

© 2026, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)

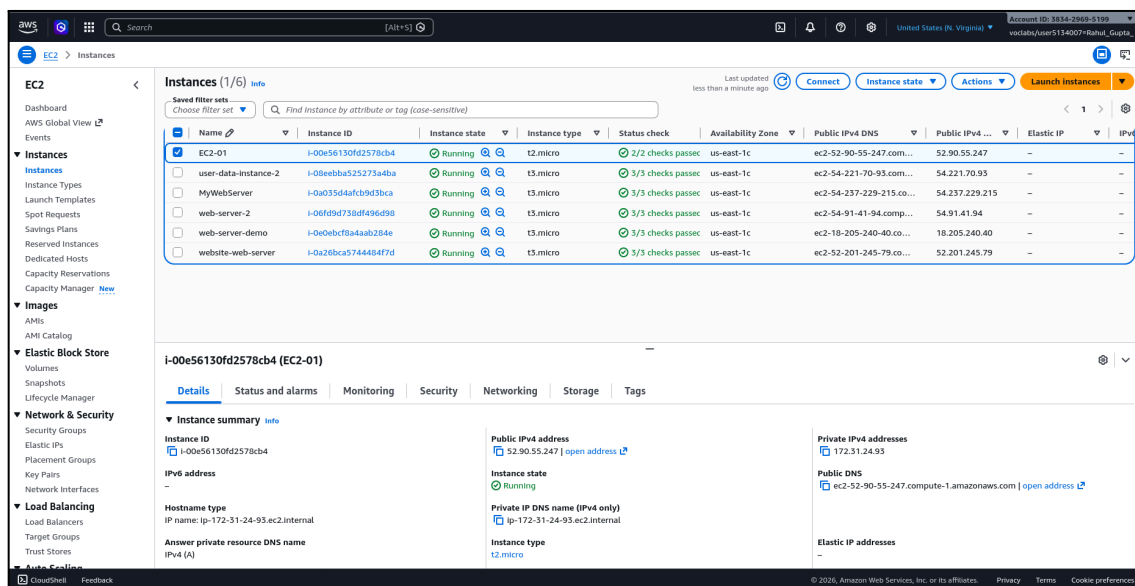
11. Click on **View all instances** to go back to your EC2 dashboard.



The screenshot shows the AWS Management Console for the EC2 service. The left sidebar contains navigation links for EC2, Images, Elastic Block Store, Network & Security, and Load Balancing. The main content area displays a table of instances. The table has columns for Name, Instance ID, Instance state, Instance type, Status check, Availability Zone, Public IPv4 DNS, Public IPv4 address, Elastic IP, and IPv6. The instances listed are EC2-01, user-data-instance-2, MyWebServer, web-server-2, web-server-demo, and website-web-server. All instances are in the 'Running' state. Below the table, there is a 'Select an instance' dropdown menu.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6
EC2-01	i-00e56130fd2578cb4	Pending	t2.micro	–	us-east-1c	ec2-52-90-55-247.com...	52.90.55.247	–	–
user-data-instance-2	i-08eebba525273a4ba	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-54-221-70-93.com...	54.221.70.93	–	–
MyWebServer	i-0a035d4afcb9d3bca	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-54-237-229-215.co...	54.237.229.215	–	–
web-server-2	i-06fd9d738df496d98	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-54-91-41-94.comp...	54.91.41.94	–	–
web-server-demo	i-0e0ebcf8a4aab284e	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-18-205-240-40.co...	18.205.240.40	–	–
website-web-server	i-0a26bca57444847fd	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-52-201-245-79.co...	52.201.245.79	–	–

12. Wait 2 to 3 minutes for the instance state to show **Running** and status checks to complete.



The screenshot shows the AWS Management Console for the EC2 service, displaying the details of instance EC2-01. The instance is in the 'Running' state. The 'Instance summary' section shows the instance ID, public IPv4 address, private IPv4 addresses, public DNS, and elastic IP addresses. The 'Status and alarms' section shows the instance state as 'Running' and the status checks as '3/3 checks passed'.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6
EC2-01	i-00e56130fd2578cb4	Running	t2.micro	2/2 checks passed	us-east-1c	ec2-52-90-55-247.com...	52.90.55.247	–	–
user-data-instance-2	i-08eebba525273a4ba	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-54-221-70-93.com...	54.221.70.93	–	–
MyWebServer	i-0a035d4afcb9d3bca	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-54-237-229-215.co...	54.237.229.215	–	–
web-server-2	i-06fd9d738df496d98	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-54-91-41-94.comp...	54.91.41.94	–	–
web-server-demo	i-0e0ebcf8a4aab284e	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-18-205-240-40.co...	18.205.240.40	–	–
website-web-server	i-0a26bca57444847fd	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-52-201-245-79.co...	52.201.245.79	–	–

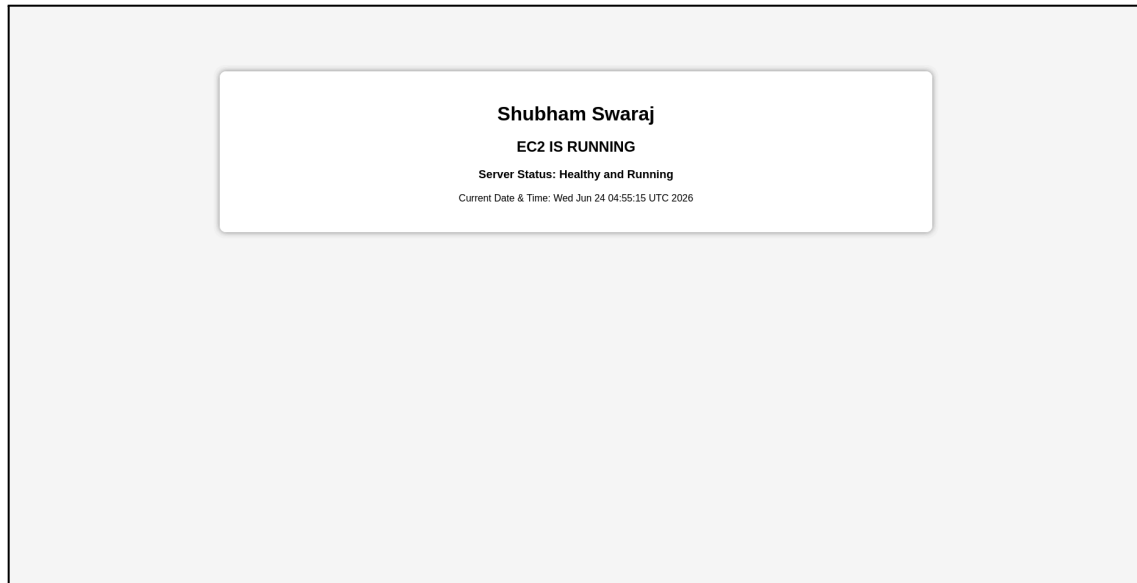
Instance summary

Instance ID	Public IPv4 address	Private IPv4 addresses	Public DNS	Elastic IP addresses
i-00e56130fd2578cb4	52.90.55.247 open address	172.31.24.93	ec2-52-90-55-247.compute-1.amazonaws.com open address	–

Instance state

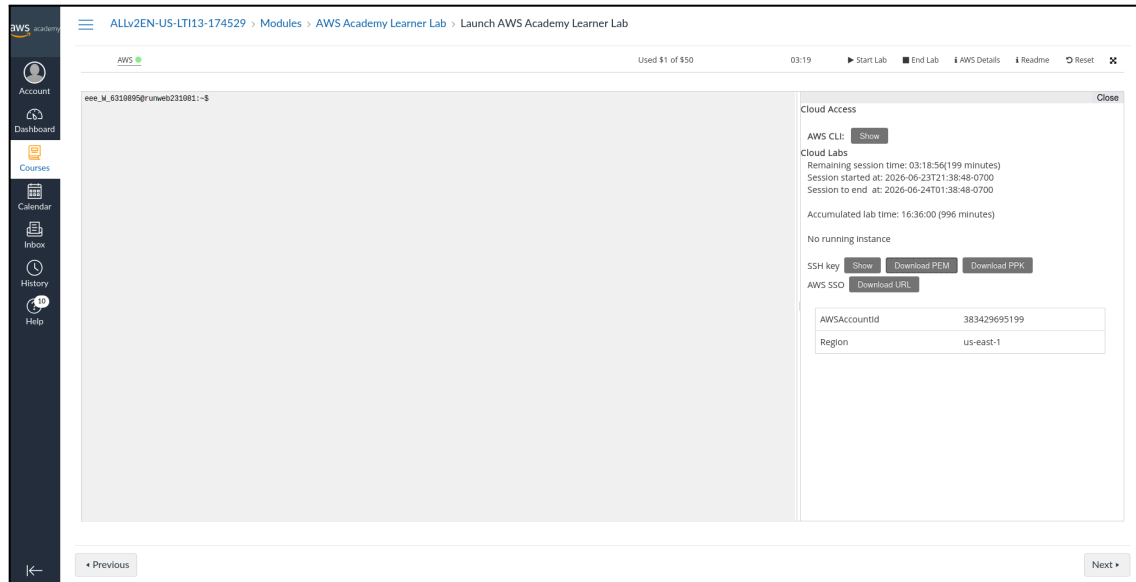
Instance state	Private IP DNS name (IPv4 only)	Instance type
Running	ip-172-31-24-93.ec2.internal	t2.micro

13. Select **EC2-01** from the list and look at the **Details** tab at the bottom to find your **Public IPv4 address**.
- Copy that Public IP address, open a new tab in your web browser, paste the IP address into the URL bar (make sure it uses **http://** and not **https://**), and press Enter.
 - Verify that your custom webpage appears showing "Shubham Swaraj", "Healthy and Running", and the timestamp.



Part 2: Connect to Ubuntu and Execute Programming Environment

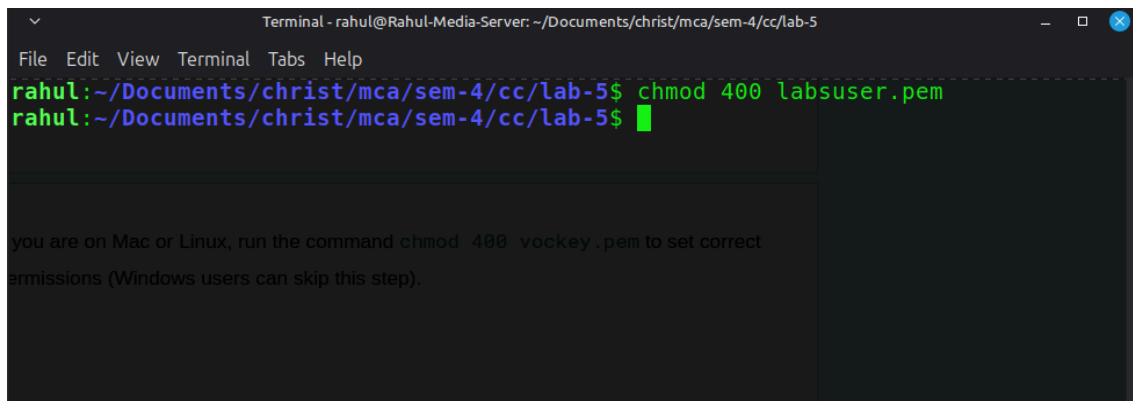
1. Go back to your AWS Learner Lab console window (where you started the lab) and click on the **AWS Details** button.



2. Click on **Download CLI credentials** or click **Download PEM** to get your private key file if it isn't automatically saved as **vokeykey.pem**.
3. Move your downloaded **vokeykey.pem** file to a known folder on your local computer (e.g., your Downloads or Desktop folder).
4. Open your local terminal (Mac/Linux) or Command Prompt/Git Bash (Windows).
5. Navigate to the folder containing your key using the change directory command, for example: **cd Downloads**

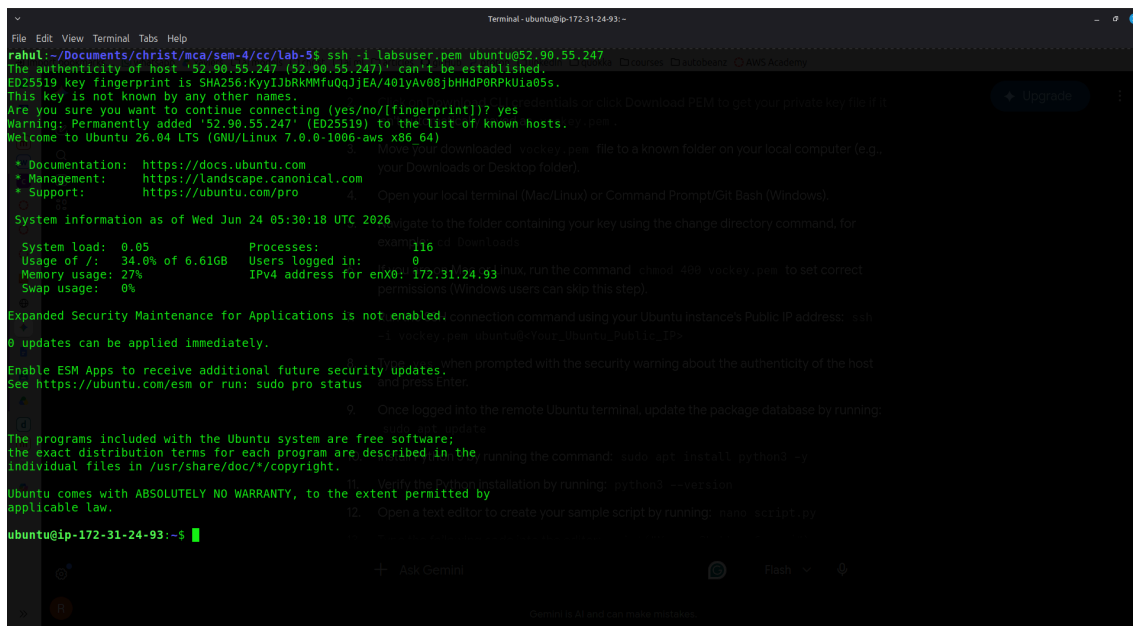


6. If you are on Mac or Linux, run the command `chmod 400 vockey.pem` to set correct permissions (Windows users can skip this step).



```
Terminal - rahul@Rahul-Media-Server: ~/Documents/christ/mca/sem-4/cc/lab-5
File Edit View Terminal Tabs Help
rahul:~/Documents/christ/mca/sem-4/cc/lab-5$ chmod 400 labsuser.pem
rahul:~/Documents/christ/mca/sem-4/cc/lab-5$
```

7. Run the SSH connection command using your Ubuntu instance's Public IP address:
- ```
ssh -i vockey.pem ubuntu@<Your_Ubuntu_Public_IP>
```
8. Type `yes` when prompted by the security warning about the authenticity of the host, then press Enter.



```
Terminal - ubuntu@ip-172-31-24-93
File Edit View Terminal Tabs Help
rahul:~/Documents/christ/mca/sem-4/cc/lab-5$ ssh -i labsuser.pem ubuntu@52.90.55.247
The authenticity of host '52.90.55.247 (52.90.55.247)' can't be established.
ED25519 key fingerprint is SHA256:KyYIJbRkMMfuQjJEA/401yAv08jbHdP0RPkUia05s.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '52.90.55.247' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation: https://docs.ubuntu.com
 * Management: https://landscape.canonical.com
 * Support: https://ubuntu.com/pro

System information as of Wed Jun 24 05:30:18 UTC 2026
System load: 0.05 Processes: 116
Usage of /: 34.0% of 6.61GB Users logged in: 0
Memory usage: 27% IPv4 address for enx0: 172.31.24.93
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-24-93:~$
```

9. Once logged into the remote Ubuntu terminal, update the package database by running: `sudo apt update`

```
ubuntu@ip-172-31-24-93:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-24-93:~$
```

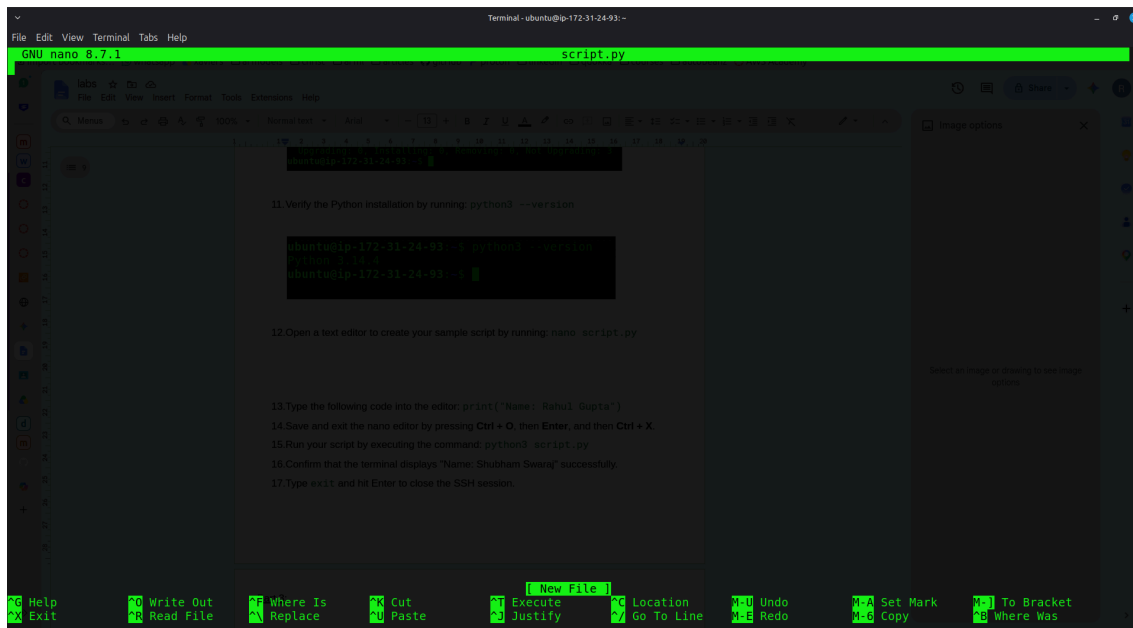
10. Install Python 3 by running the command: `sudo apt install python3 -y`

```
ubuntu@ip-172-31-24-93:~$ sudo apt install python3 -y
python3 is already the newest version (3.14.3-0ubuntu2).
python3 set to manually installed.
Summary:
 Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 3
ubuntu@ip-172-31-24-93:~$
```

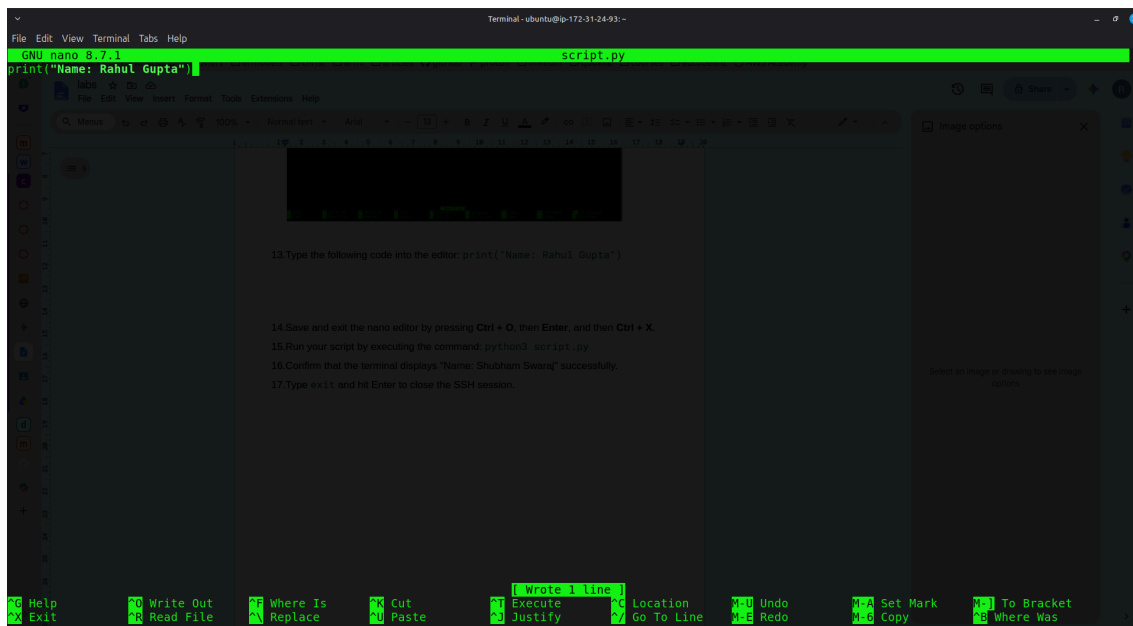
11. Verify the Python installation by running: `python3 --version`

```
ubuntu@ip-172-31-24-93:~$ python3 --version
Python 3.14.4
ubuntu@ip-172-31-24-93:~$
```

12. Open a text editor to create your sample script by running: `nano script.py`



13. Type the following code into the editor: `print("Name: Rahul Gupta")`



14. Save and exit the nano editor by pressing **Ctrl + O**, then **Enter**, and then **Ctrl + X**.

15. Run your script by executing the command: `python3 script.py`
16. Confirm that the terminal displays "Name: Rahul Gupta" successfully.

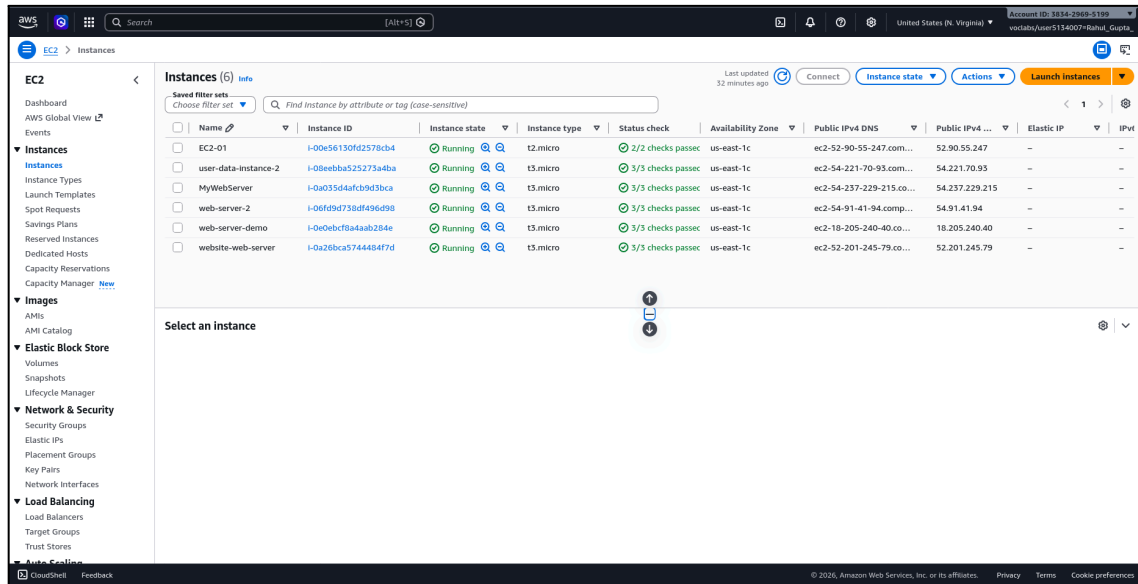
```
ubuntu@ip-172-31-24-93:~$ python3 script.py
Name: Rahul Gupta
ubuntu@ip-172-31-24-93:~$
```

17. Type `exit` and hit Enter to close the SSH session.

```
ubuntu@ip-172-31-24-93:~$ exit
logout
Connection to 52.90.55.247 closed.
rahul:~/Documents/christ/mca/sem-4/cc/lab-5$
```

## Part 3: Launch and Verify Windows Server

1. Go back to your AWS EC2 Management Console page in your browser.



2. Click on **Instances** on the left menu, and then click **Launch instance**.
3. Name this second instance by typing **EC2-02** in the **Name** field.

**Name and tags** [Info](#)

**Name**

[Add additional tags](#)

4. Under **Application and OS Images (Amazon Machine Image)**, select **Windows**.

5. Ensure the default **Microsoft Windows Server 2025 Base (or 2022 Base) - Free Tier Eligible** AMI is selected.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

My AMIs

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Microsoft Windows Server 2025 Base

Free tier eligible

ami-09639480113b0df96 (64-bit (x86))

Virtualization: hvm    ENA enabled: true    Root device type: ebs

Description

Microsoft Windows 2025 Datacenter edition. [English]

Microsoft Windows Server 2025 Full Locale English AMI provided by Amazon

| Architecture | AMI ID                | Publish Date | Username      |                              |
|--------------|-----------------------|--------------|---------------|------------------------------|
| 64-bit (x86) | ami-09639480113b0df96 | 2026-06-10   | Administrator | <div>Verified provider</div> |

6. Under **Instance type**, keep it as **t3.micro**

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3    2 vCPU    1 GiB Memory    Current generation: true    On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour    On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour    On-Demand RHEL base pricing: 0.0392 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

7. Under **Key pair (login)**, select the exact same **vockey** key pair from the dropdown menu.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vockey

▼

[Create new key pair](#)

For Windows instances, you use a key pair to decrypt the administrator password. You then use the decrypted password to connect to your instance.



8. Under **Network settings**, leave **Create security group** checked.
  - a. Ensure that **Allow RDP traffic from (Anywhere 0.0.0.0/0)** is checked by default to allow remote desktop access.

▼ Network settings Info

VPC - required Info

vpc-0033feb051601c6f172.31.0.0/16(default)↻

Subnet Info

No preference↕↻ Create new subnet ↗

Availability Zone Info

No preference↕↻ Enable additional zones ↗

Auto-assign public IP Info

Enable↕

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

launch-wizard-8

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and \_-:/()#,@!+=&:~\*.

Description - required Info

launch-wizard-8 created 2026-06-24T05:39:08.148Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 3389, 0.0.0.0/0) Remove

| Type Info        | Protocol Info                                                               | Port range Info                       |
|------------------|-----------------------------------------------------------------------------|---------------------------------------|
| rdp              | TCP                                                                         | 3389                                  |
| Source type Info | Source Info                                                                 | Description - optional Info           |
| Anywhere         | <div>Q Add CIDR, prefix list or security group</div> <div>0.0.0.0/0 ✕</div> | <div>e.g. SSH for admin desktop</div> |

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

✕

Add security group rule

9. Click the **Launch instance** button on the right side panel.

▼ Summary

Number of instances Info

1↕

Software Image (AMI)

Microsoft Windows Server 2025 ...read more

ami-09639480113b0df96

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

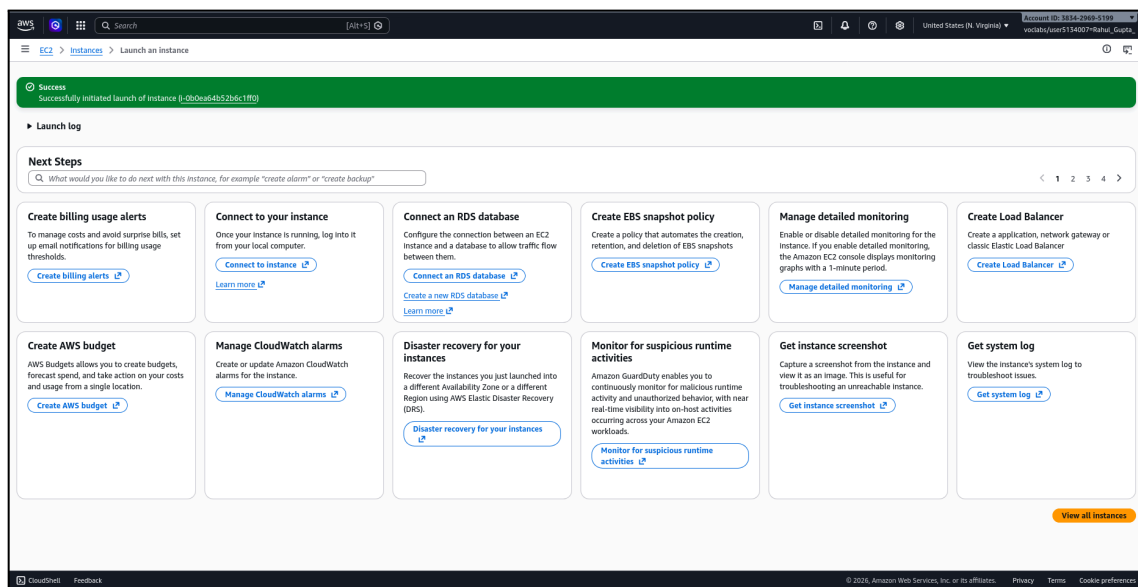
Storage (volumes)

1 volume(s) - 30 GiB

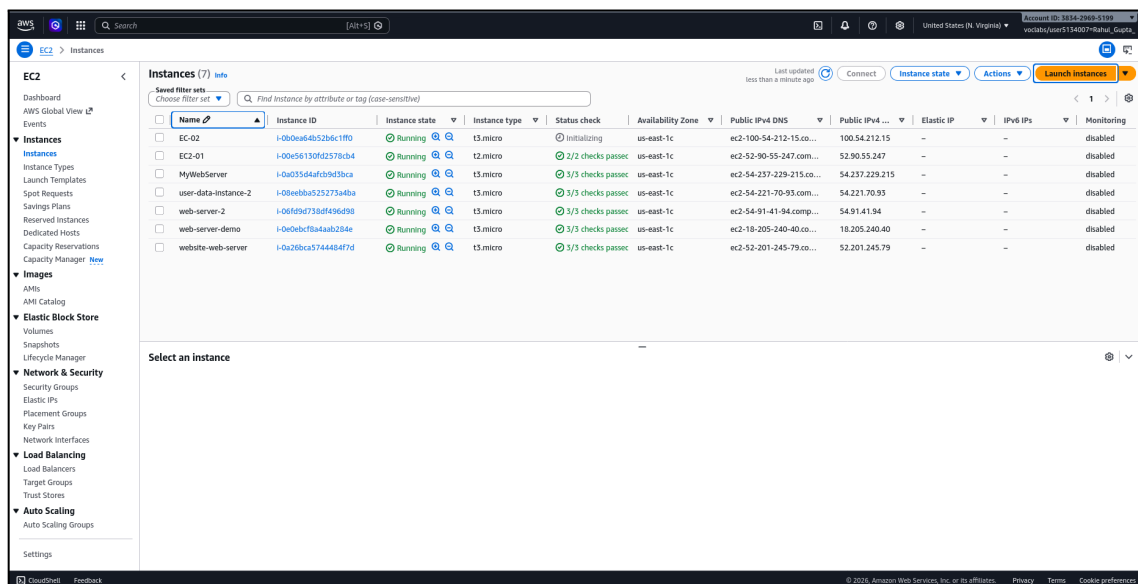
Cancel

Launch instance

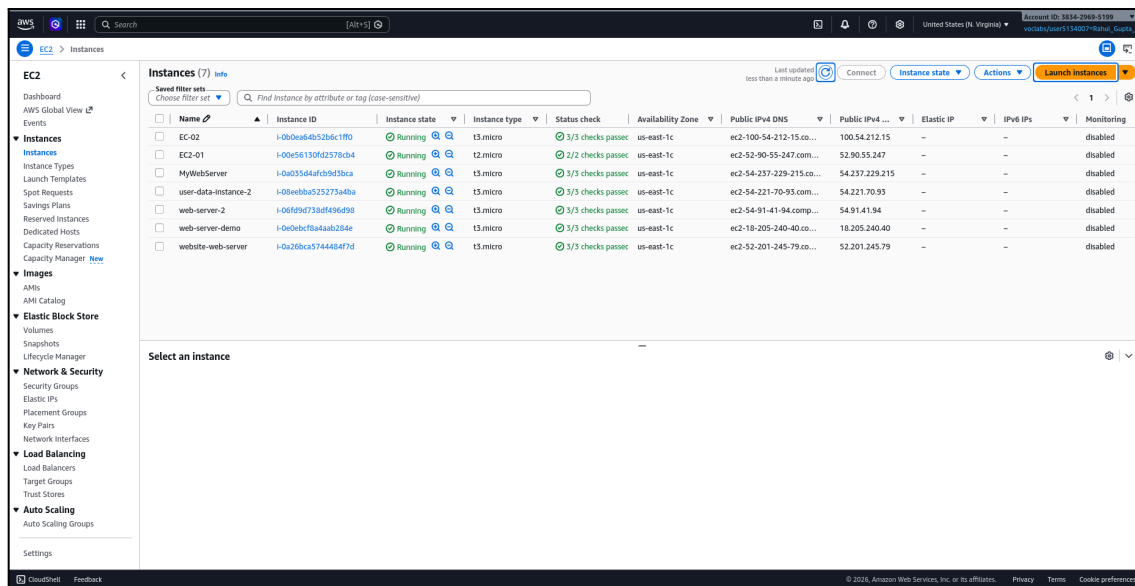
Preview code



10. Click on **View all instances** at the bottom of the success screen.



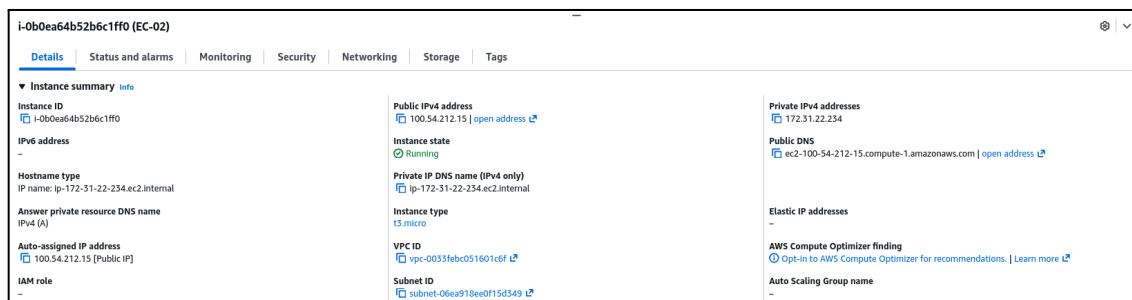
11. Wait for the **Instance state** of **EC2-02** to transition to **Running**.



12. Click on the checkbox next to **EC2-02** to view its specific configurations.

13. Verify and record the **Instance State** from the details pane, confirming it says **Running**.

14. Locate and copy the **Public IPv4 address** from the details pane to verify network assignment.



15. Scroll down further in the Details tab to view **Platform** and **AMI name** to verify the Operating System details match Microsoft Windows Server.

|                                                                              |                                                                                                    |                                                                                         |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| i-0b0ea64b52b6c1ff0 (EC-02)                                                  |                                                                                                    |                                                                                         |
| ▼ Instance details <a href="#">Info</a>                                      |                                                                                                    |                                                                                         |
| AMI ID<br><a href="#">ami-09639400113b0df96</a>                              | Monitoring<br>disabled                                                                             | Platform details<br><a href="#">Windows</a>                                             |
| AMI name<br><a href="#">Windows_Server-2025-English-Full-Base-2026.06.10</a> | Allowed image<br>-                                                                                 | Termination protection<br>Disabled                                                      |
| Stop protection<br>Disabled                                                  | Launch time<br><a href="#">Wed Jun 24 2026 11:12:32 GMT+0530 (India Standard Time)</a> (4 minutes) | AMI location<br><a href="#">amazon/Windows_Server-2025-English-Full-Base-2026.06.10</a> |
| Instance reboot migration<br>Default (On)                                    | Instance auto-recovery<br>Default                                                                  | Lifecycle<br>normal                                                                     |
| Stop-hibernate behavior<br>Disabled                                          | AMI Launch index<br>0                                                                              | Key pair assigned at launch<br><a href="#">vockey</a>                                   |
| State transition reason<br>-                                                 | Credit specification<br>unlimited                                                                  | Kernel ID<br>-                                                                          |
| State transition message<br>-                                                | Usage operation<br><a href="#">RunInstances:0002</a>                                               | RAM disk ID<br>-                                                                        |